

7.5.2 DRVCTRL Register in STEP/DIR Mode

DRVCTRL		Driver Control in STEP/DIR Mode (SDOFF=0)	
Bit	Name	Function	Comment
19	0	Register address bit	
18	0	Register address bit	
17	0	Reserved	
16	0	Reserved	
15	0	Reserved	
14	0	Reserved	
13	0	Reserved	
12	0	Reserved	
11	0	Reserved	
10	0	Reserved	
9	INTPOL	Enable STEP interpolation	0: Disable STEP pulse interpolation. 1: Enable STEP pulse multiplication by 16.
8	DEDGE	Enable double edge STEP pulses	0: Rising STEP pulse edge is active, falling edge is inactive. 1: Both rising and falling STEP pulse edges are active.
7	0	Reserved	
6	0	Reserved	
5	0	Reserved	
4	0	Reserved	
3	MRES3	Microstep resolution for STEP/DIR mode	Microsteps per 90°: %0000: 256 %0001: 128 %0010: 64 %0011: 32 %0100: 16 %0101: 8 %0110: 4 %0111: 2 (halfstep) %1000: 1 (fullstep)
2	MRES2		
1	MRES1		
0	MRES0		

7.6 Chopper Control Register (CHOPCONF)

CHOPCONF		Chopper Configuration	
Bit	Name	Function	Comment
19	1	Register address bit	
18	0	Register address bit	
17	0	Register address bit	
16	TBL1	Blanking time	Blanking time interval, in system clock periods: %00: 16 %01: 24 %10: 36 %11: 54
15	TBL0		
14	CHM	Chopper mode	This mode bit affects the interpretation of the HDEC, HEND, and HSTRT parameters shown below. 0 Standard mode (SpreadCycle) 1 Constant t_{OFF} with fast decay time. Fast decay time is also terminated when the negative nominal current is reached. Fast decay is after on time.
13	RNDTF	Random TOFF time	Enable randomizing the slow decay phase duration: 0: Chopper off time is fixed as set by bits t_{OFF} 1: Random mode, t_{OFF} is random modulated by $dN_{CLK} = -24 \dots +6$ clocks.
12	HDEC1	Hysteresis decrement interval or Fast decay mode	CHM=0 Hysteresis decrement period setting, in system clock periods: %00: 16 %01: 32 %10: 48 %11: 64 CHM=1 HDEC1=0: current comparator can terminate the fast decay phase before timer expires. HDEC1=1: only the timer terminates the fast decay phase. HDEC0: MSB of fast decay time setting.
11	HDEC0		
10	HEND3	Hysteresis end (low) value or Sine wave offset	CHM=0 %0000 ... %1111: Hysteresis is -3, -2, -1, 0, 1, ..., 12 (1/512 of this setting adds to current setting) This is the hysteresis value which becomes used for the hysteresis chopper.
9	HEND2		
8	HEND1		CHM=1 %0000 ... %1111: Offset is -3, -2, -1, 0, 1, ..., 12 This is the sine wave offset and 1/512 of the value becomes added to the absolute value of each sine wave entry.
7	HEND0		
6	HSTRT2	Hysteresis start value or Fast decay time setting	CHM=0 Hysteresis start offset from HEND: %000: 1 %100: 5 %001: 2 %101: 6 %010: 3 %110: 7 %011: 4 %111: 8 <i>Effective: HEND+HSTRT must be ≤ 15</i>
5	HSTRT1		
4	HSTRT0		CHM=1 Three least-significant bits of the duration of the fast decay phase. The MSB is HDEC0. Fast decay time is a multiple of system clock periods: $N_{CLK} = 32 \times (HDEC0 + HSTRT)$

CHOPCONF		Chopper Configuration	
Bit	Name	Function	Comment
3	TOFF3	Off time/MOSFET disable	Duration of slow decay phase. If TOFF is 0, the MOSFETs are shut off. If TOFF is nonzero, slow decay time is a multiple of system clock periods: $N_{CLK} = 24 + (32 \times TOFF)$ %0000: Driver disable, all bridges off %0001: 1 (use with TBL of minimum 24 clocks) %0010 ... %1111: 2 ... 15
2	TOFF2		
1	TOFF1		
0	TOFF0		

7.7 CoolStep Control Register (SMARTEN)

SMARTEN		CoolStep Configuration	
Bit	Name	Function	Comment
19	1	Register address bit	
18	0	Register address bit	
17	1	Register address bit	
16	0	Reserved	
15	SEIMIN	Minimum CoolStep current	0: ½ CS current setting 1: ¼ CS current setting
14	SEDN1	Current decrement speed	Number of times that the StallGuard2 value must be sampled equal to or above the upper threshold for each decrement of the coil current: %00: 32 %01: 8 %10: 2 %11: 1
13	SEDN0		
12	0	Reserved	
11	SEMAX3	Upper CoolStep threshold as an offset from the lower threshold	If the StallGuard2 measurement value SG is sampled equal to or above (SEMIN+SEMAX+1) x 32 enough times, then the coil current scaling factor is decremented.
10	SEMAX2		
9	SEMAX1		
8	SEMAX0		
7	0	Reserved	
6	SEUP1	Current increment size	Number of current increment steps for each time that the StallGuard2 value SG is sampled below the lower threshold: %00: 1 %01: 2 %10: 4 %11: 8
5	SEUP0		
4	0	Reserved	
3	SEMIN3	Lower CoolStep threshold/CoolStep disable	If SEMIN is 0, CoolStep is disabled. If SEMIN is nonzero and the StallGuard2 value SG falls below SEMIN x 32, the CoolStep current scaling factor is increased.
2	SEMIN2		
1	SEMIN1		
0	SEMIN0		

7.8 StallGuard2 Control Register (SGCSCONF)

SGCSCONF		StallGuard2™ and Current Setting	
Bit	Name	Function	Comment
19	1	Register address bit	
18	1	Register address bit	
17	0	Register address bit	
16	SFILT	StallGuard2 filter enable	0: Standard mode, fastest response time. 1: Filtered mode, updated once for each four fullsteps to compensate for variation in motor construction, highest accuracy.
15	0	Reserved	
14	SGT6	StallGuard2 threshold value	The StallGuard2 threshold value controls the optimum measurement range for readout and stall indicator output (SG_TST). A lower value results in a higher sensitivity and less torque is required to indicate a stall. The value is a two's complement signed integer. Range: -64 to +63
13	SGT5		
12	SGT4		
11	SGT3		
10	SGT2		
9	SGT1		
8	SGT0		
7	0	Reserved	
6	0	Reserved	
5	0	Reserved	
4	CS4	Current scale (scales digital currents A and B)	Current scaling for SPI and step/direction operation. %00000 ... %11111: 1/32, 2/32, 3/32, ... 32/32 This value is biased by 1 and divided by 32, so the range is 1/32 to 32/32. Example: CS=20 is 21/32 current
3	CS3		
2	CS2		
1	CS1		
0	CS0		

7.9 Driver Control Register (DRVCONF)

DRVCONF		Driver Configuration	
Bit	Name	Function	Comment
19	1	Register address bit	
18	1	Register address bit	
17	1	Register address bit	
16	TST	Reserved TEST mode	Must be cleared for normal operation. When set, the SG_TST output exposes digital test values, and the TEST_ANA output exposes analog test values.
15	SLPH1	Slope control, high side	%00: Minimum %01: Minimum (+tc) %10: Medium (+tc) %11: Maximum Temperature compensated mode (tc) increases the high-side MOSFET gate driver strength if the overtemperature warning temperature is reached. This compensates for temperature dependency of high-side slope control.
14	SLPH0		
13	SLPL1	Slope control, low side	
12	SLPL0		
11	0	Reserved	Set to 0
10	DISS2G	Short to GND protection disable	0: Short to GND protection is enabled. 1: Short to GND protection is disabled.
9	TS2G1	Short to GND detection timer	%00: 3.2µs. %01: 1.6µs. %10: 1.2µs. %11: 0.8µs.
8	TS2G0		
7	SDOFF	STEP/DIR interface disable	0: Enable STEP and DIR interface. 1: Disable STEP and DIR interface. SPI interface is used to move motor.
6	VSENSE	Sense resistor voltage-based current scaling	0: Full-scale sense resistor voltage is 310mV. 1: Full-scale sense resistor voltage is 165mV. (Full-scale refers to a current setting of 31 and a DAC value of 255.)
5	RDSEL1	Select value for read out (RD bits)	%00 Microstep position read back
4	RDSEL0		%01 StallGuard2 level read back
			%10 StallGuard2 & CoolStep current level read back
			%11 *) All status flags and detectors
3	OTSENS *)	Overtemperature sensitivity	0: Shutdown at 150°C 1: Sensitive shutdown at 136°C
2	SHRTSENS *)	Short to GND detection sensitivity	0: Low sensitivity 1: High sensitivity – better protection for high side FETs
1	EN_PFD *)	Enable Passive fast decay / 5V undervoltage threshold	0: No additional motor dampening. 1: Motor dampening to reduce motor resonance at medium velocity. In addition, this bit reduces the lower nominal operation voltage limit from 7V to 4.5V
0	EN_S2VS *)	Enable short to VS & CLK fail protection	0: Short to VS and clock failsafe protection disabled 1: Short to VS / overcurrent protection enabled. In addition, enables protection against clock input CLK fail, when using an external clock source.

*) These bits have a function for TMC2660C only. Setting these bits / functions for TMC2660 does not have any effect. The TMC2660 and TMC2660C behave identically with setting 0.

7.10 Read Response

For every write command sent to the motor driver, a 20-bit response is returned to the motion controller. The response has one of three formats, as selected by the RDSEL parameter in the DRVCONF register. The table below shows these formats. Software must not depend on the value of any bit shown as reserved.

any bit shown as reserved.

DRVSTATUS					Read Response	
Bit	Name				Function	Comment
	RDSEL					
	%00	%01	%10	%11		
19	MSTEP9	SG9	SG9	UV_7V	Microstep counter / StallGuard2 SG9:0 / StallGuard2 SG9:5 and CoolStep SE4:0 / Diagnostic status	Microstep position in sine table for coil A in STEP/DIR mode. MSTEP9 is the Polarity bit: 0: Current flows from OA1 pins to OA2 pins. 1: Current flows from OA2 pins to OA1 pins.
18	MSTEP8	SG8	SG8	ENN in		
17	MSTEP7	SG7	SG7	S2VSB		
16	MSTEP6	SG6	SG6	S2GB		
15	MSTEP5	SG5	SG5	S2VSA		
14	MSTEP4	SG4	SE4	S2GA		
13	MSTEP3	SG3	SE3	OT150		
12	MSTEP2	SG2	SE2	OT136		
11	MSTEP1	SG1	SE1	OT120		
10	MSTEP0	SG0	SE0	OT100		
9	0			1	-	%11 response allows to distinguish -C type. Non-C-type delivers %00 in each case.
8	0			1		
7	STST				Standstill indicator	0: No standstill condition detected. 1: No active edge occurred on the STEP input during the last 2 ²⁰ system clock cycles.
6	OLB				Open load indicator	0: No open load condition detected. 1: No chopper event has happened during the last period with constant coil polarity. Only a current above 1/16 of the maximum setting can clear this bit! <i>Hint:</i> This bit is only a status indicator. The chip takes no other action when this bit is set. False indications may occur during fast motion and at standstill. Check this bit only during slow motion.
5	OLA					
4	SHORTB				Short detection status	0: No short condition. 1: Short condition. The short counter is incremented by each short circuit and the chopper cycle is suspended. The counter is decremented for each phase polarity change. The MOSFETs are shut off when the counter reaches 3 and remain shut off until the shutdown condition is cleared by disabling and re-enabling the driver. The shutdown condition becomes reset by de-asserting the ENN input or clearing the TOFF parameter.
3	SHORTA					
2	OTPW				Overtemp. warning	0: No overtemperature warning condition. 1: Warning threshold is active.
1	OT				Overtemp. shutdown	0: No overtemperature shutdown condition. 1: Overtemperature shutdown has occurred.
0	SG				StallGuard2 status	0: No motor stall detected. 1: StallGuard2 threshold has been reached, and the SG TST output is driven high.

7.11 Device Initialization

The following sequence of SPI commands is an example of enabling the driver and initializing the chopper:

```
SPI = $901B4;    // Hysteresis mode
or
SPI = $94557;    // Constant toff mode

SPI = $D001F;    // Current setting: $d001F (max. current)

SPI = $E0010;    // low driver strength, StallGuard2 read, SDOFF=0

SPI = $00000;    // 256 microstep setting
```

First test of CoolStep current control:

```
SPI = $A8202;    // Enable CoolStep with minimum current ¼ CS
```

The configuration parameters should be tuned to the motor and application for optimum performance.

8 STEP/DIR Interface

The STEP and DIR inputs provide a simple, standard interface compatible with many existing motion controllers. The MicroPlyer STEP pulse interpolator brings the smooth motor operation of high-resolution microstepping to applications originally designed for coarser stepping and reduces pulse bandwidth.

8.1 Timing

Figure 8.1 shows the timing parameters for the STEP and DIR signals, and the table below gives their specifications. When the DEDGE mode bit in the DRVCTRL register is set, both edges of STEP are active. If DEDGE is cleared, only rising edges are active. STEP and DIR are sampled and synchronized to the system clock. An internal analog filter removes glitches on the signals, such as those caused by long PCB traces. If the signal source is far from the chip, and especially if the signals are carried on cables, the signals should be filtered or differentially transmitted.

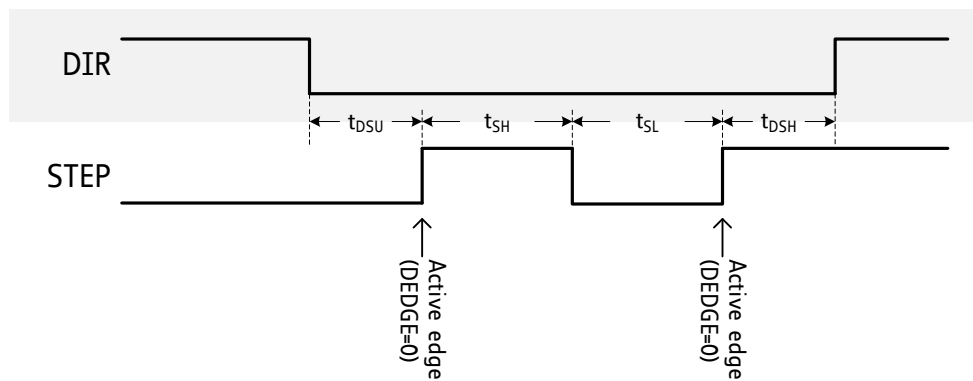


Figure 8.1 STEP and DIR timing.

STEP and DIR Interface Timing		AC-Characteristics				
		clock period is t_{CLK}				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Step frequency (at maximum microstep resolution)	f_{STEP}	DEEDGE=0			$\frac{1}{2} f_{CLK}$	
		DEEDGE=1			$\frac{1}{4} f_{CLK}$	
Fullstep frequency	f_{FS}				$f_{CLK}/512$	
STEP input low time	t_{SL}		$\max(t_{FILTSD}, t_{CLK}+20)$			ns
STEP input high time	t_{SH}		$\max(t_{FILTSD}, t_{CLK}+20)$			ns
DIR to STEP setup time	t_{DSU}		20			ns
DIR after STEP hold time	t_{DSH}		20			ns
STEP and DIR spike filtering time TMC2660C *)	t_{FILTSD}	Rising and falling edge	12	20	40	ns
STEP and DIR sampling relative to rising CLK input	$t_{SDCLKHI}$	Before rising edge of CLK		t_{FILTSD}		ns

*) The TMC2660C has reduced filter time to reduce the danger of skipped step pulses

8.2 Microstep Table

The internal microstep table maps the sine function from 0° to 90°. Symmetries allow mapping the sine and cosine functions from 0° to 360° with this table. The angle is encoded as a 10-bit unsigned integer MSTEP provided by the microstep counter. The size of the increment applied to the counter while microstepping through this table is controlled by the microstep resolution setting MRES in the DRVCTRL register. Depending on the DIR input, the microstep counter is increased (DIR=0) or decreased (DIR=1) by the step size with each STEP active edge. Despite many entries in the last quarter of the table being equal, the electrical angle continuously changes, because either the sine wave or cosine wave is in an area, where the current vector changes monotonically from position to position. Figure 8.2 shows the table. The largest values are 248, which leaves headroom used for adding an offset.

Entry	0-31	32-63	64-95	96-127	128-159	160-191	192-223	224-255
0	1	49	96	138	176	207	229	243
1	2	51	97	140	177	207	230	244
2	4	52	98	141	178	208	231	244
3	5	54	100	142	179	209	231	244
4	7	55	101	143	180	210	232	244
5	8	57	103	145	181	211	232	245
6	10	58	104	146	182	212	233	245
7	11	60	105	147	183	212	233	245
8	13	61	107	148	184	213	234	245
9	14	62	108	150	185	214	234	246
10	16	64	109	151	186	215	235	246
11	17	65	111	152	187	215	235	246
12	19	67	112	153	188	216	236	246
13	21	68	114	154	189	217	236	246
14	22	70	115	156	190	218	237	247
15	24	71	116	157	191	218	237	247
16	25	73	118	158	192	219	238	247
17	27	74	119	159	193	220	238	247
18	28	76	120	160	194	220	238	247
19	30	77	122	161	195	221	239	247
20	31	79	123	163	196	222	239	247
21	33	80	124	164	197	223	240	247
22	34	81	126	165	198	223	240	248
23	36	83	127	166	199	224	240	248
24	37	84	128	167	200	225	241	248
25	39	86	129	168	201	225	241	248
26	40	87	131	169	201	226	241	248
27	42	89	132	170	202	226	242	248
28	43	90	133	172	203	227	242	248
29	45	91	135	173	204	228	242	248
30	46	93	136	174	205	228	243	248
31	48	94	137	175	206	229	243	248

Figure 8.2 Internal microstep table showing the first quarter of the sine wave.

8.3 Changing Resolution

The application may need to change the microstepping resolution to get the best performance from the motor. For example, high-resolution microstepping may be used for precision operations on a workpiece, and then fullstepping may be used for maximum torque at maximum velocity to advance to the next workpiece. When changing to coarse resolutions like fullstepping or halfstepping, switching should occur at or near positions that correspond to steps in the lower resolution, as shown in Table 8.1.

Step Position	MSTEP Value	Coil A Current	Coil B Current
Half step 0	0	0%	100%
Full step 0	128	70.7%	70.7%
Half step 1	256	100%	0%
Full step 1	384	70.7%	-70.7%
Half step 2	512	0%	-100%
Full step 2	640	-70.7%	-70.7%
Half step 3	768	-100%	0%
Full step 3	896	-70.7%	70.7%

Table 8.1 Optimum positions for changing to halfstep and fullstep resolution

8.4 MicroPlyer Step Interpolator

For each active edge on STEP, MicroPlyer produces 16 microsteps at 256x resolution, as shown in Figure 8.3. MicroPlyer is enabled by setting the INTPOL bit in the DRVCTRL register. It supports input at 16x resolution, which it transforms into 256x resolution. The step rate for each 16 microsteps is determined by measuring the time interval of the previous step period and dividing it into 16 equal parts. The maximum time between two microsteps corresponds to 2^{20} (roughly one million system clock cycles), for an even distribution of 1/256 microsteps. At 16MHz system clock frequency, this results in a minimum step input frequency of 16Hz for MicroPlyer operation (one fullstep per second). A lower step rate causes the STST bit to be set, which indicates a standstill event. At that frequency, microsteps occur at a rate of $\frac{\text{system clock frequency}}{2^{16}} \approx 250\text{Hz} = 244$.

Attention

MicroPlyer only works well with a stable STEP frequency. Do not use the DEDGE option if the STEP signal does not have a 50% duty cycle.

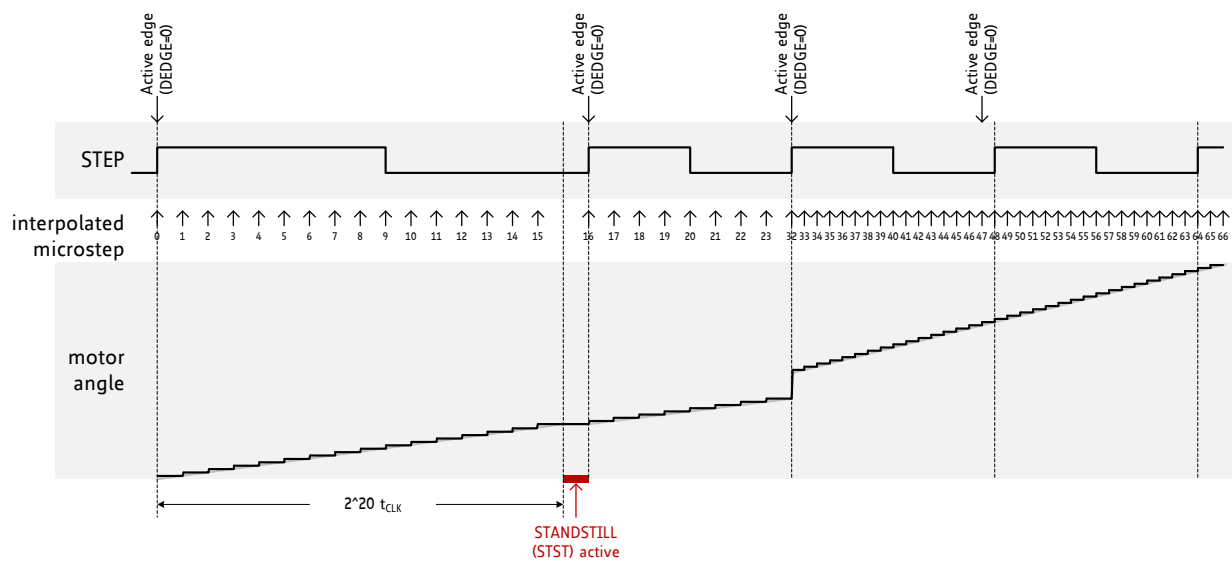


Figure 8.3 MicroPlyer microstep interpolation with rising STEP frequency.

In Figure 8.3, the first STEP cycle is long enough to set the STST bit. This bit is cleared on the next STEP active edge. Then, the STEP frequency increases and after one cycle at the higher rate MicroPlyer increases the interpolated microstep rate. During the last cycle at the slower rate, MicroPlyer did not generate all 16 microsteps, so there is a small jump in motor angle between the first and second cycles at the higher rate.

8.5 Standstill current reduction

When a standstill event is detected, the motor current should be reduced to save energy and reduce heat dissipation in the power MOSFET stage. This is especially true at halfstep positions, which are a worst-case condition for the driver and motor because the full energy is consumed in one bridge and one motor coil.

Attention:

Stand still current reduction is required when operating near the thermal limits of the application. Refer the stand still current limitations in chapter 18 as a guideline.

Hint

Implement a current reduction to 10% to 75% of the required run current for motor standstill. This saves more than 50% to more than 90% of energy. The actual level depends on the required holding force and on the required microstep precision during standstill. In standalone mode, a reduction to 50% current is possible using a configuration input.

9 Current Setting

The internal 5V supply voltage available at the pin 5VOUT is used as a reference for the coil current regulation based on the sense resistor voltage measurement. The desired maximum motor current is set by selecting an appropriate value for the sense resistor. The sense resistor voltage range can be selected by the VSENSE bit in the DRVCONF register. The low sensitivity (high sense resistor voltage, VSENSE=0) brings best and most robust current regulation, while high sensitivity (low sense resistor voltage; VSENSE=1) reduces power dissipation in the sense resistor. This setting reduces the power dissipation in the sense resistor by nearly half.

After choosing the VSENSE setting and selecting the sense resistor, the currents to both coils are scaled by the 5-bit current scale parameter CS in the SGCSCONF register. The sense resistor value is chosen so that the maximum desired current (or slightly more) flows at the maximum current setting (CS = %11111).

Using the internal sine wave table, which has amplitude of 248, the RMS motor current can be calculated by:

$$I_{RMS} = \frac{CS + 1}{32} * \frac{V_{FS}}{R_{SENSE}} * \frac{1}{\sqrt{2}}$$

The momentary motor current is calculated as:

$$I_{MOT} = \frac{CURRENT_{A/B}}{248} * \frac{CS + 1}{32} * \frac{V_{FS}}{R_{SENSE}}$$

where:

CS is the effective current scale setting as set by the CS bits and modified by CoolStep. The effective value ranges from 0 to 31.

V_{FS} is the sense resistor voltage at full scale, as selected by the VSENSE control bit (refer to the electrical characteristics).

$CURRENT_{A/B}$ is the value set by the current setting in SPI mode or the internal sine table in STEP/DIR mode.

Parameter	Description	Setting	Comment
CS	<i>Current scale.</i> Scales both coil current values as taken from the internal sine wave table or from the SPI interface. For high precision motor operation, work with a current scaling factor in the range 16 to 31, because scaling down the current values reduces the effective microstep resolution by making microsteps coarser. This setting also controls the maximum current value set by CoolStep.	0 ... 31	Scaling factor: 1/32, 2/32, ... 32/32
VSENSE	Allows control of the sense resistor <i>voltage range</i> or adaptation of one electronic module to different maximum motor currents.	0	325mV
		1	173mV

Pay special attention concerning the thermal design and current limits imposed by it!
Be sure to reduce the current to a value at or below the maximum standstill current limits as specified in chapter 18. This is important due to thermal restrictions for continuous current in a single bridge. The worst case is standstill in a half step position where one bridge has 0 current, and the other bridge has *RMS value * $\sqrt{2}$ current*.

9.1 Sense Resistors

Sense resistors should be carefully selected. The full motor current flows through the sense resistors. As they also see the switching spikes from the MOSFET bridges, a low-inductance type such as film or composition resistors is required to prevent spikes causing ringing. A low-inductance, low-resistance PCB layout is essential. Keep the high-current interconnections as short as possible as shown in Figure 9.1. A massive ground plane is best. Because the sense resistor inputs are susceptible to damage from negative overvoltages, an additional input protection resistor helps protect against a motor cable break or ringing on long motor cables.

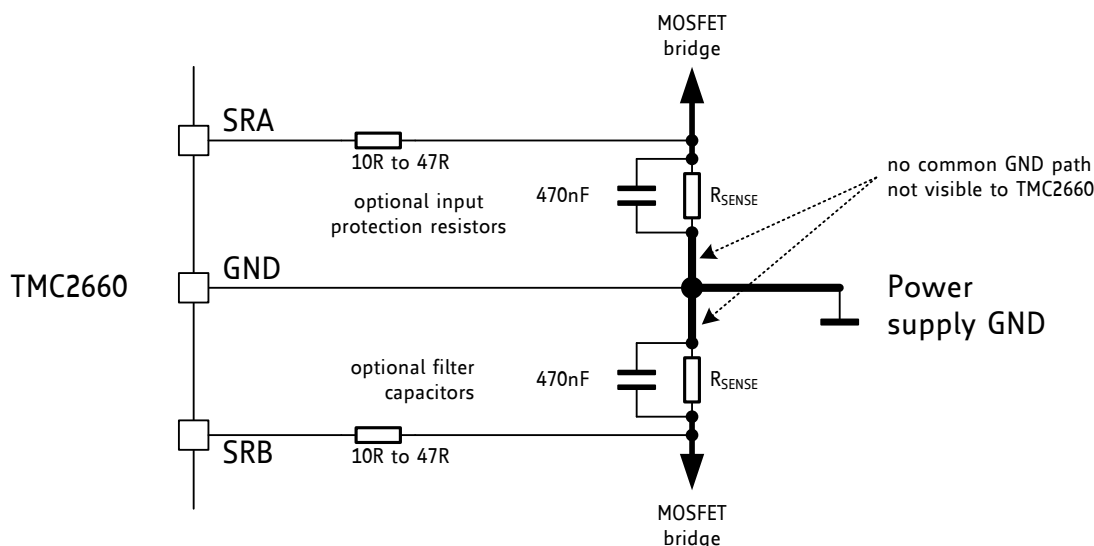


Figure 9.1 Sense resistor grounding and protection components

The sense resistor needs to be able to conduct the peak motor coil current, especially in motor standstill conditions, unless standby power is reduced. Under normal conditions, the sense resistor sees the motor coil current with a duty cycle well below 50%.

The peak sense resistor power dissipation is:

$$P_{RSMAX} = \frac{\left(V_{SENSE} * \frac{CS + 1}{32}\right)^2}{R_{SENSE}}$$

For high-current applications, power dissipation is halved by using the lower sense resistor voltage setting and the corresponding lower resistance value. In this case, any voltage drop in the PCB traces has a larger influence on the result. A compact power stage layout with massive ground plane is best to avoid parasitic resistance effects.

CHOICE OF R_{SENSE} AND RESULTING MAX. MOTOR CURRENT FOR $CS=31$		
R_{SENSE} [Ω]	RMS current [A] $V_{SENSE}=0$	RMS current [A] $V_{SENSE}=1$
0.33	0.7	0.35
0.27	0.8	0.43
0.22	1.0	0.53
0.15	1.5	0.8
0.12	1.8	1.0
0.10	2.2	1.2

10 Chopper Operation

The currents through both motor coils are controlled using choppers. The choppers work independently of each other. Figure 10.1 shows the three chopper phases:

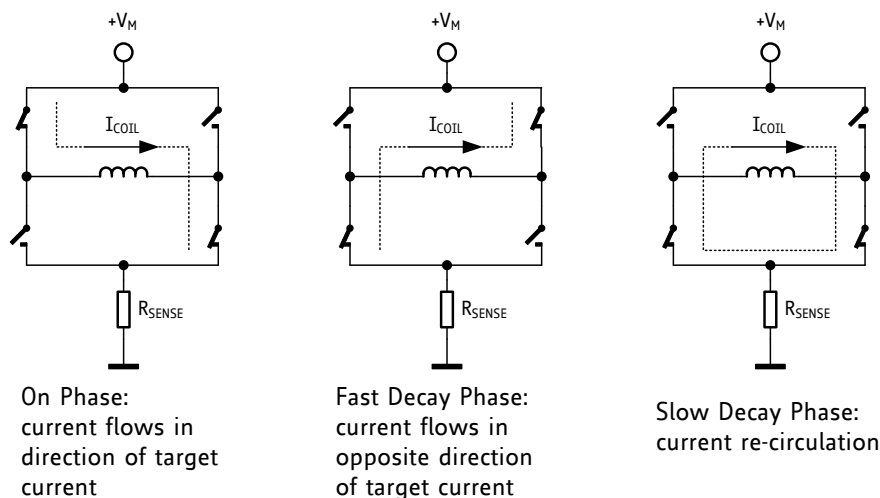


Figure 10.1 Chopper phases.

Although the current could be regulated using only on phases and fast decay phases, insertion of the slow decay phase is important to reduce electrical losses and current ripple in the motor. The duration of the slow decay phase is specified in a control parameter and sets an upper limit on the chopper frequency. The current comparator can measure coil current during phases when the current flows through the sense resistor, but not during the slow decay phase, so the slow decay phase is terminated by a timer. The on phase is terminated by the comparator when the current through the coil reaches the target current. The fast decay phase may be terminated by either the comparator or another timer.

When the coil current is switched, spikes at the sense resistors occur due to charging and discharging parasitic capacitances. During this time, typically one or two microseconds, the current cannot be measured. Blanking is the time when the input to the comparator is masked to block these spikes.

There are two chopper modes available: a new high-performance chopper algorithm called SpreadCycle and a proven constant off-time chopper mode. The constant off-time mode cycles through three phases: on, fast decay, and slow decay. The SpreadCycle mode cycles through four phases: on, slow decay, fast decay, and a second slow decay.

These parameters are used for controlling both chopper modes:

Parameter	Description	Setting	Comment
TOFF	Off time. This setting controls the duration of the slow decay time and limits the maximum chopper frequency. For most applications an off time within the range of 5µs to 20µs will fit. If the value is 0, the MOSFETs are all shut off and the motor can freewheel. If the value is 1 to 15, the number of system clock cycles in the slow decay phase is: $N_{CLK} = (TOFF \cdot 32) + 24$ The SD-Time is $t = \frac{1}{f_{CLK}} \cdot N_{CLK}$	0 1... 15	Chopper off. Off time setting. A setting in the range of 2-5 is recommended for SpreadCycle, higher values for classic chopper.

Parameter	Description	Setting	Comment
TBL	Blanking time. This time needs to cover the switching event and the duration of the ringing on the sense resistor. For most low-current applications, a setting of 16 or 24 is good. For high-current applications, a setting of 36 or 54 may be required.	0	16 system clock cycles
		1	24 system clock cycles
		2	36 system clock cycles
		3	54 system clock cycles
CHM	Chopper mode bit. SpreadCycle is recommended for most applications.	0	SpreadCycle mode
		1	Constant off time mode
EN_PFD	Enable passive fast decay. Setting this bit, adds a passive fast decay phase of 512 clock cycles for each bridge following zero crossing of the motor current. The passive fast decay will dampen the energy of motor resonances at medium velocity.	1	Resonance dampening on.

10.1 SpreadCycle Chopper

The SpreadCycle (patented) chopper algorithm is a precise and simple to use chopper mode which automatically determines the optimum length for the fast-decay phase. The SpreadCycle will provide superior microstepping quality even with default settings. Several parameters are available to optimize the chopper to the application.

Each chopper cycle is comprised of an on phase, a slow decay phase, a fast decay phase and a second slow decay phase (see Figure 10.3). The two slow decay phases and the two blank times per chopper cycle put an upper limit to the chopper frequency. The slow decay phases typically make up for about 30%-70% of the chopper cycle in standstill and are important for low motor and driver power dissipation.

Calculation of a starting value for the slow decay time *TOFF*:

EXAMPLE:

Target Chopper frequency: 25kHz.

Assumption: Two slow decay cycles make up for 50% of overall chopper cycle time

$$t_{OFF} = \frac{1}{25kHz} * \frac{50}{100} * \frac{1}{2} = 10\mu s$$

For the *TOFF* setting this means:

$$TOFF = (t_{OFF} * f_{CLK} - 24)/32$$

With 12 MHz clock this gives a setting of *TOFF*=3.0, i.e. 3.

With 16 MHz clock this gives a setting of *TOFF*=4.25, i.e. 4 or 5.

The hysteresis start setting forces the driver to introduce a minimum amount of current ripple into the motor coils. The current ripple must be higher than the current ripple which is caused by resistive losses in the motor to give best microstepping results. This will allow the chopper to precisely regulate the current both for rising and for falling target current. The time required to introduce the current ripple into the motor coil also reduces the chopper frequency. Therefore, a higher hysteresis setting will lead to a lower chopper frequency. The motor inductance limits the ability of the chopper to follow a changing motor current. Further the duration of the on phase and the fast decay must be longer than the blanking time because the current comparator is disabled during blanking.

It is easiest to find the best setting by starting from a low hysteresis setting (e.g. *HSTRT*=0, *HEND*=0) and increasing *HSTRT*, until the motor runs smoothly at low velocity settings. This can best be checked when measuring the motor current either with a current probe or by probing the sense resistor voltages (see Figure 10.2). Checking the sine wave shape near zero transition will show a

small ledge between both half waves in case the hysteresis setting is too small. At medium velocities (i.e. 100 to 400 fullsteps per second), a too low hysteresis setting will lead to increased humming and vibration of the motor.

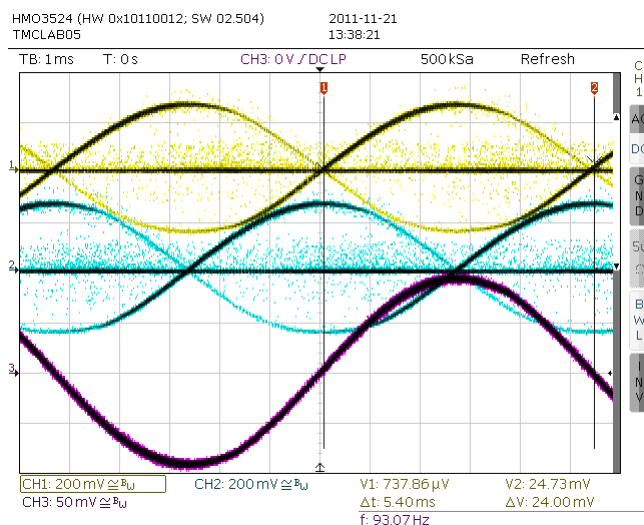


Figure 10.2 No ledges in current wave with sufficient hysteresis (magenta: current A, yellow & blue: sense resistor voltages A and B)

A too high hysteresis setting will lead to reduced chopper frequency and increased chopper noise but will not yield any benefit for the wave shape.

Quick Start

For detail tuning procedure see Application Note AN001 - *Parameterization of SpreadCycle*

As experiments show, the setting is quite independent of the motor, because higher current motors typically also have a lower coil resistance. Choosing a medium default value for the hysteresis (for example, effective HSTRT+HEND=10) normally fits most applications. The setting can be optimized by experimenting with the motor: A too low setting will result in reduced microstep accuracy, while a too high setting will lead to more chopper noise and motor power dissipation. When measuring the sense resistor voltage in motor standstill at a medium coil current with an oscilloscope, a too low setting shows a fast decay phase not longer than the blanking time. When the fast decay time becomes slightly longer than the blanking time, the setting is optimum. You can reduce the off-time setting, if this is hard to reach.

The hysteresis principle could in some cases lead to the chopper frequency becoming too low, for example when the coil resistance is high compared to the supply voltage. This is avoided by splitting the hysteresis setting into a start setting (HSTRT+HEND) and an end setting (HEND). An automatic hysteresis decremter (HDEC) interpolates between these settings, by decrementing the hysteresis value stepwise each 16, 32, 48, or 64 system clock cycles. At the beginning of each chopper cycle, the hysteresis begins with a value which is the sum of the start and the end values (HSTRT+HEND), and decrements during the cycle, until either the chopper cycle ends, or the hysteresis end value (HEND) is reached. This way, the chopper frequency is stabilized at high amplitudes and low supply voltage situations, if the frequency gets too low. This avoids the frequency reaching the audible range.

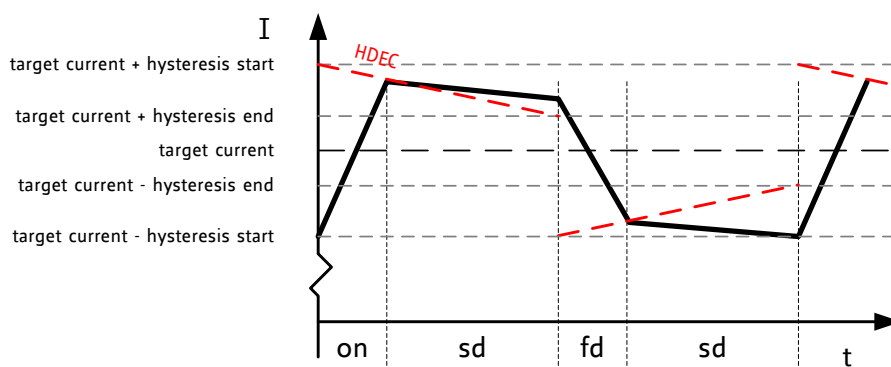


Figure 10.3 SpreadCycle chopper mode showing the coil current during a chopper cycle

Three parameters control SpreadCycle mode:

Parameter	Description	Setting	Comment
HSTRT	<i>Hysteresis start</i> setting. Please remark, that this value is an offset to the hysteresis end value HEND.	0... 7	This setting adds to HEND. %000: 1 %100: 5 %001: 2 %101: 6 %010: 3 %110: 7 %011: 4 %111: 8
HEND	<i>Hysteresis end</i> setting. Sets the hysteresis end value after a number of decrements. Decrement interval time is controlled by HDEC. The sum HSTRT+HEND must be <16. At a current setting CS of max. 30 (amplitude reduced to 240), the sum is not limited.	0... 2	Negative HEND: -3... -1 %0000: -3 %0001: -2 %0010: -1
		3	Zero HEND: 0 %0011: 0
		4... 15	Positive HEND: 1... 12 %0100: 1 %1010: 7 %0101: 2 %1011: 8 %0110: 3 %1100: 9 %0111: 4 %1101: 10 %1000: 5 %1110: 11 %1001: 6 %1111: 12
HDEC	<i>Hysteresis decrement</i> setting. This setting determines the slope of the hysteresis during on time and during fast decay time. It sets the number of system clocks for each decrement.	0... 3	0: fast decrement 3: very slow decrement %00: 16 %01: 32 %10: 48 %11: 64

EXAMPLE:

A hysteresis of 4 has been chosen. You might decide to not use hysteresis decrement. In this case set:

HEND=6 (sets an effective end value of 6-3=3)
HSTRT=0 (sets minimum hysteresis, i.e. 1: 3+1=4)

In order to take advantage of the variable hysteresis, we can set most of the value to the HSTRT, i.e. 4, and the remaining 1 to hysteresis end. The resulting configuration register values are as follows:

HEND=0 (sets an effective end value of -3)
HSTRT=6 (sets an effective start value of hysteresis end +7: 7-3=4)

Hint

Highest motor velocities benefit from setting TOFF to 2 or 3 and a short TBL of 2 or 1.

10.2 Constant Off-Time Mode

The classic constant off-time chopper uses a fixed-time fast decay following each on phase. While the duration of the on phase is determined by the chopper comparator, the fast decay time needs to be fast enough for the driver to follow the falling slope of the sine wave, but it should not be so long that it causes excess motor current ripple and power dissipation. This can be tuned using an oscilloscope or evaluating motor smoothness at different velocities. A good starting value is a fast decay time setting similar to the slow decay time setting.

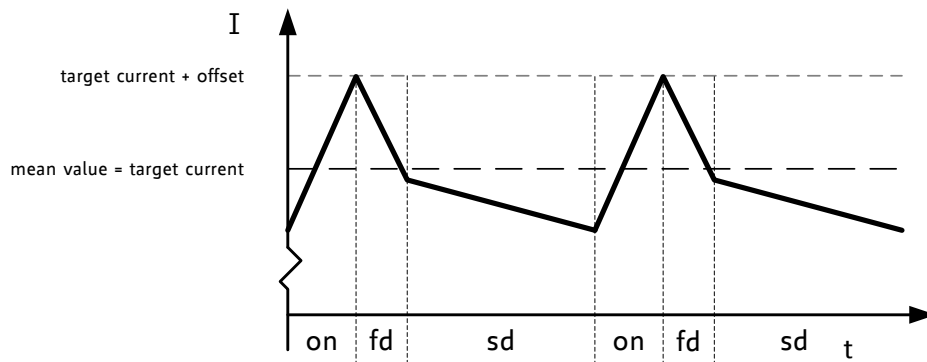


Figure 10.4 Constant off-time chopper with offset showing the coil current during two cycles

After tuning the fast decay time, the offset should be tuned for a smooth zero crossing. This is necessary because the fast decay phase makes the absolute value of the motor current lower than the target current (see Figure 10.5). If the zero offset is too low, the motor stands still for a short moment during current zero crossing. If it is set too high, it makes a larger microstep. Typically, a positive offset setting is required for smoothest operation.

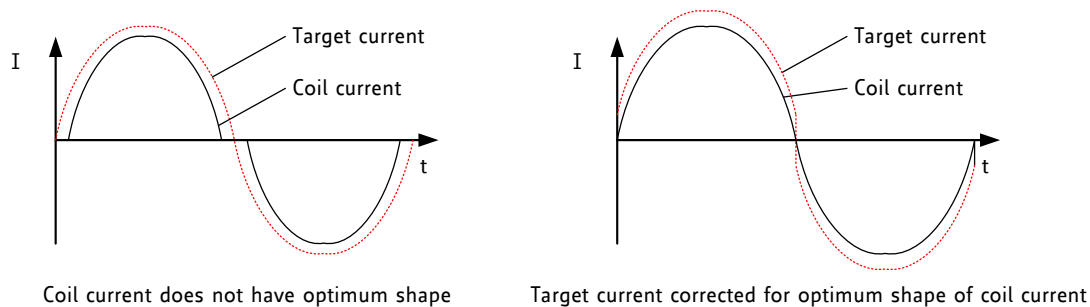


Figure 10.5 Zero crossing with correction using sine wave offset.

Three parameters control constant off-time mode:

Parameter	Description	Setting	Comment
TFD (HSTART & HDEC0)	<i>Fast decay time</i> setting. With CHM=1, these bits control the portion of fast decay for each chopper cycle.	0	Slow decay only.
		1... 15	Duration of fast decay phase.
OFFSET (HEND)	<i>Sine wave offset</i> . With CHM=1, these bits control the sine wave offset. A positive offset corrects for zero crossing error.	0...2	Negative offset: -3... -1
		3	No offset: 0
		4... 15	Positive offset: 1... 12
NCCFD (HDEC1)	Selects usage of the <i>current comparator</i> for termination of the <i>fast decay</i> cycle. If current comparator is enabled, it terminates the fast decay cycle in case the current reaches a higher negative value than the actual positive value.	0	Enable comparator termination of fast decay cycle.
		1	End by time only.

10.2.1 Random Off Time

In the constant off-time chopper mode, both coil choppers run freely without synchronization. The frequency of each chopper mainly depends on the coil current and the motor coil inductance. The inductance varies with the microstep position. With some motors, a slightly audible beat can occur between the chopper frequencies when they are close together. This typically occurs at a few microstep positions within each quarter wave. This effect is usually not audible when compared to mechanical noise generated by ball bearings, etc. Another factor which can cause a similar effect is a poor layout of the sense resistor GND connections.

Hint

A common cause of motor noise is a bad PCB layout causing coupling of both sense resistor voltages. Noise caused by an insufficient PCB layout cannot be overcome by chopper settings.

To minimize the effect of a beat between both chopper frequencies, an internal random generator is provided. It modulates the slow decay time setting when switched on by the RNDTF bit. The RNDTF feature further spreads the chopper spectrum, reducing electromagnetic emission on single frequencies.

Parameter	Description	Setting	Comment
RNDTF	Enables a random off-time generator, which slightly modulates the off time t_{OFF} using a random polynomial.	0	Disable.
		1	Random modulation enable.

11 Power MOSFET Stage

The gate current for the power MOSFETs can be adapted to influence the slew rate at the coil outputs. The main features of the stage are:

- 5V gate drive voltage for low-side N-MOS transistors, 10V for high-side P-MOS transistors.
- The gate drivers protect the bridges actively against cross-conduction using an internal Q_{GD} protection that holds the MOSFETs safely off.
- Automatic break-before-make logic minimizes dead time and diode-conduction time.
- Integrated overcurrent protection detects a short of the motor wires and protects the MOSFETs.

The low-side gate driver is supplied by the 5VOUT pin. The low-side driver supplies 0V to the MOSFET gate to close the MOSFET, and 5VOUT to open it. The high-side gate driver voltage is supplied by the VS and the VHS pin. VHS is more negative than VS and allows opening the VS referenced high-side MOSFET. The high-side driver supplies VS to the P channel MOSFET gate to close the MOSFET and VHS to open it. The effective low-side gate voltage is roughly 5V; the effective high-side gate voltage is roughly 8V.

Parameter	Description	Setting	Comment
SLPL	Low-side slope control. Controls the MOSFET gate driver current. Set to 0, 1 or 2. Use fastest slope to minimize package power dissipation.	0... 3	%00: Minimum. %01: Minimum. %10: Medium. %11: Maximum.
SLPH	High-side slope control. Controls the MOSFET gate driver current. For the TMC2660 set identical to SLPL to match LS slope.	0... 3	%00: Minimum. %01: Minimum+TC. %10: Medium+TC. %11: Maximum.

11.1 Break-Before-Make Logic

Each half-bridge has to be protected against cross-conduction during switching events. When switching off the low-side MOSFET, its gate first needs to be discharged before the high-side MOSFET is allowed to switch on. The same goes when switching off the high-side MOSFET and switching on the low-side MOSFET. The time for charging and discharging of the MOSFET gates depends on the MOSFET gate charge and the gate driver current set by SLPL and SLPH. The BBM (break-before-make) logic measures the gate voltage and automatically delays turning on the opposite bridge transistor until its counterpart is discharged. This way, the bridge will always switch with optimized timing independent of the slope setting.

11.2 ENN Input

The MOSFETs can be completely disabled in hardware by pulling the ENN input high. This allows the motor to free-wheel. An equivalent function can be performed in software by setting the parameter TOFF to zero. The hardware disable is available for allowing the motor to be hot plugged. For the TMC2660, it can be used in overvoltage situations. The TMC2660 can withstand voltages of up to 60V when the MOSFETs are disabled. If a hardware disable function is not needed, tie ENN low.